

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			H ₂ O ₂			1	1		
	(b)	(i)		40 ±1		1		1	1	
		(ii)		lead oxide is the best catalyst – credit for reason any of following • it produces oxygen at a faster rate than the other catalysts • it produces more oxygen in 120s than the other catalysts • it gives a steeper curve than the other catalysts • it reacts fastest			1	1		1
		(iii)		the same mass of all catalysts is left over ✓	1			1		1

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					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		mesh has greater surface area (1) more collisions (1)	2			2		2
		(ii)		they cause global warming ✓			1	1		
		(iii)		more nitrogen oxides are converted than carbon monoxide up to 100°C ✓			1	1		
		(iv)		award (1) for statement of opinion with basic reason e.g. not very effective because not all harmful gases are converted or effective because it removes most of the harmful gases award additional (1) for further detail from passage/data e.g. - it takes 30 minutes for a catalytic converter to work effectively - harmful gases can still escape in the first 30 minutes of a journey - catalytic converters are not effective for short journeys / journeys that take less than 30 minutes - pollutant gases / carbon monoxide and nitrogen oxides will not be converted into harmless gases at low temperatures - after 30 minutes up to 60-70% conversion of carbon monoxide and nitrogen oxides into “safe” gases			2	2		
				Question 5 total	3	1	6	10	1	4